

Anuranan Bharadwaj

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EDUCATION

Embry-Riddle Aeronautical University

Daytona Beach, FL

Bachelor of Science in Aerospace Engineering - Jet Propulsion

May 2026

Minor in Computer Science

GPA: 3.8

Courses: Jet Propulsion | Structures | Materials | Aerodynamics | Thermodynamics | Stability & Control | OOP | DSA

EXPERIENCE

Turbomachinery Software Intern

January – April 2026

Concepts NREC

White River Junction, VT

- Designed and implemented **Python automation frameworks** using **AxCent's APIs** to execute automated testing of turbomachinery blade design workflows, replacing manual validation and generating structured **Excel-based verification reports**.
- Developed **automated validation pipelines** for **gas turbine performance** modeling tools, testing AxCent outputs across **diverse blade configurations** and accelerating **QA cycles** for proprietary turbomachinery design software.
- Analyzing blade profile repositories** applied to 3D turbine rotor and stator models, validating key geometric parameters including **blade count**, **leading/trailing edge radius**, **gauge angle**, **inlet angle**, and **rotation direction (CW/CCW)** to ensure design consistency and reliability.

Undergraduate Research Assistant

May 2025 – April 2026

Gas Turbine Lab – Embry-Riddle Aeronautical University

Daytona Beach, FL

- Developed a **Python-Excel** tool to compute **turbine blade throat openings** and **deviation angles** via **Aungier's correlations** and **Mach-dependent thermodynamic cycle effects**, supporting gas turbine thermal performance analysis across operating conditions.
- Validated computational component analysis results against **experimental data**; automated **multi-stage turbine performance analysis** to accelerate **design iteration** and **aerodynamic characterization**.
- Designed and **CAD-modeled** a custom pitot rake for **multi-point total pressure measurement** in **turbine cascade rig**; supports field performance **data acquisition** and **instrumentation upgrades** for engine cycle validation.

Systems Engineering Intern

July – Aug 2024

ABH Software

Assam, India

- Engineered automated data reporting pipelines using **SQL** and **Java** to simulate real-time telemetry and logistics tracking; **reduced manual report processing by 30%** and **improved accuracy by 20%** using **Agile methodologies**.
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PROJECTS

Aircraft Stability & Control Simulation | *MATLAB, Simulink, DATCOM, FlightGear*

- Built full MATLAB/Simulink flight simulation model with lookup tables for **aerodynamic derivatives** computed via USAF DATCOM; validated **longitudinal and lateral stability** under control inputs.
- Visualized flight **performance data** in FlightGear; confirmed effective pitch, roll, and yaw response.

Aircraft Wing Structural Analysis | *FEMAP, NX Nastran, Fusion 360*

- Performed FEA on full aircraft wing geometry (spars, ribs, stringers, skin) in FEMAP/NX Nastran under **aerodynamic load** conditions; **max displacement 0.15 in** with **Jacobian mesh quality > 0.6**.
 - Interpreted **stress contours** and **displacement plots** to assess **structural margins** and **component stiffness**.
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SKILLS & Development

Engineering Software: SolidWorks | CATIA | FEMAP | CFTurbo | AxCent | Simulink | MATLAB | Fusion 360

Programming Language: Java | Python | HTML | CSS | JS | C#

Developer Tools: VS Code | Visual Studio | Git | PyCharm | Jupyter | Figma | Claude Code | N8N

Frameworks & Technologies: MERN | .NET | WPF | SQL | Pandas | Matplotlib | NumPy | Excel

- Boeing Career Program (BCP) Mentee** – Class of 2025